Task 12: Design and develop a RESTful API for Product Management using Spring Boot

Aim

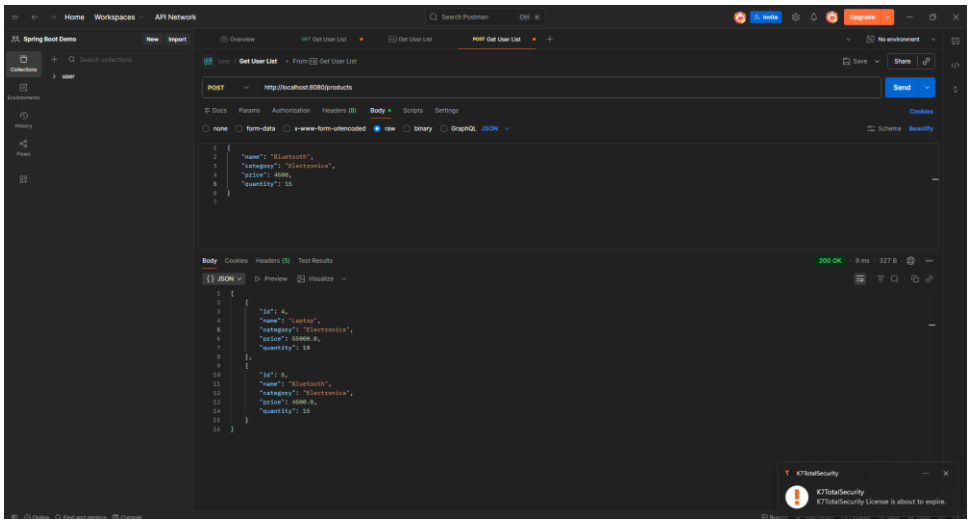
To design and develop a **RESTful API for Product Management** using **Spring Boot** that supports **GET, POST, PUT, and DELETE** operations, returns data in **JSON format**, and can be tested using a **REST client** such as Postman.

Algorithm

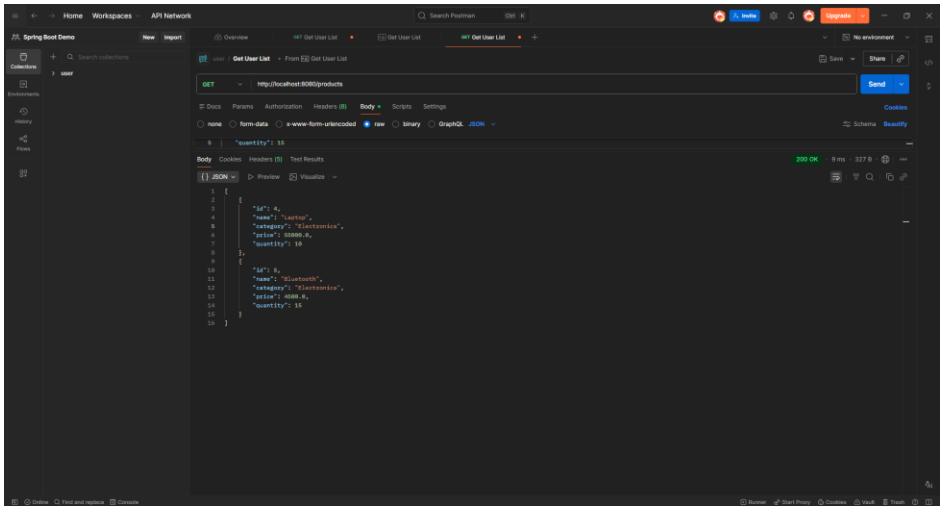
1. Start the application.
2. Create a Spring Boot project with required dependencies.
3. Configure database properties in application.properties.
4. Create a Product entity class.
5. Create a ProductRepository interface extending JpaRepository.
6. Create a ProductService class to handle business logic.
7. Create a ProductController class to expose REST endpoints.
8. Implement POST method to add a product.
9. Implement GET method to retrieve all products and retrieve product by id.
10. Implement PUT method to update an existing product.
11. Implement DELETE method to remove a product.
12. Return all responses in JSON format.
13. Run the Spring Boot application.
14. Test the API endpoints using a REST client.
15. Stop the application.

Outputs:

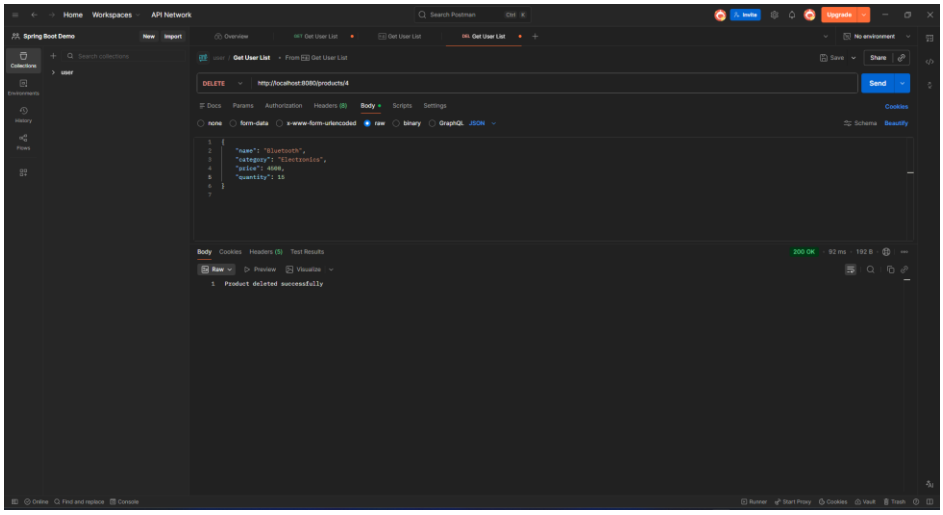
Add:



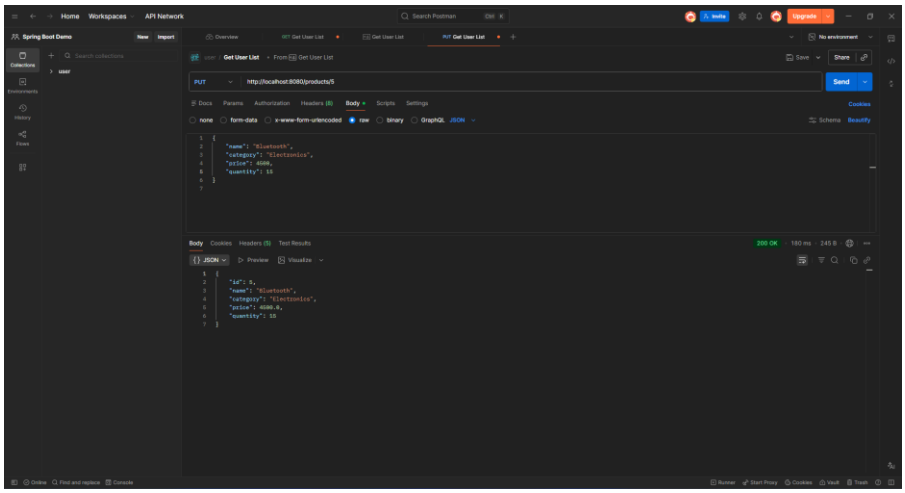
Get:



Delete:



Put:



Result:

To design and develop a **RESTful API for Product Management** using **Spring Boot** that supports **GET, POST, PUT, and DELETE** operations, returns data in **JSON format**, and can be tested using a **REST client** such as Postman is executed successfully.